Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[11pt]article acl2016 times latexsym

Instructions for ACL 2016 ProceedingsThis document has been adapted from the instructions for earlier ACL and NAACL proceedings, including those for NAACL HLT16 by Margaret Mitchell, NAACL HLT15 by Matt Post and Adam Lopez, NAACL HLT12 by Nizar Habash and William Schuler, NAACL HLT10 by Claudia Leacock and Richard Wicentowski, NAACL HLT09 by Joakim Nivre and Noah Smith, for ACL05 by Hwee Tou Ng and Kemal Oflazer, for ACL02 by Eugene

Charniak and Dekang Lin, and earlier ACL and EACL formats. Those versions were written by several people, including John Chen, Henry S. Thompson and Donald Walker. Additional elements were taken from the formatting instructions of the *International Joint Conference on Artificial Intelligence* and the *Conference on Computer Vision and Pattern Recognition.*

Author 1
XYZ Company
111 Anywhere Street
Mytown, NY 10000, USA
`author1@xyz.org` Author 2
ABC University
900 Main Street
Ourcity, PQ, Canada A1A 1T2
`author2@abc.ca`

Ringkasan

This document contains instructions for preparing ACL 2016 submissions and camera-ready manuscripts. The document itself conforms to its own specifications, and is therefore an example of what your manuscript should look like. Papers are required to conform to all the directions reported in this document. By using the provided LaTeX and BibTeX styles (`acl2016.sty`, `acl2016.bst`), the required formatting will be enabled by default.

1 Introduction

The following instructions are directed to authors of papers submitted to and accepted for publication in the ACL 2016 proceedings. All authors are required to adhere to these specifications. Authors are required to provide a Portable Document Format (PDF) version of their papers. The proceedings will be printed on A4 paper. Authors from countries where access to word-processing systems is limited should contact the publication chairs as soon as possible. Grayscale readability of all figures and graphics will be encouraged for all accepted papers (Section 2.8).

Submitted and camera-ready formatting is similar, however, the submitted paper should have:

1. Author-identifying information removed
2. A 'ruler' on the left and right margins
3. Page numbers
4. A confidentiality header.

In contrast, the camera-ready **should not have** a ruler, page numbers, nor a confidentiality header. By uncommenting `\aclfinalcopy` at the top of this document, it will compile to produce an example of the camera-ready formatting; by leaving it commented out, the document will be anonymized for initial submission. Authors should place this command after the `\usepackage` declarations when preparing their camera-ready manuscript with the ACL 2016 style.

2 General Instructions

Manuscripts must be in two-column format. Exceptions to the two-column format include the title, as well as the authors' names and complete addresses (only in the final version, not in the version submitted for review), which must be centered at the top of the first page (see the guidelines in Subsection 2.4), and any full-width figures or tables. Type single-spaced. Do not number the pages in the camera-ready version. Start all pages directly under the top margin. See the guidelines later regarding formatting the first page.

The maximum length of a manuscript is eight (8) pages for the main conference, printed single-sided, plus two (2) pages for references (see Section 3 for additional information on the maximum number of pages).

By uncommenting `\aclfinalcopy` at the top of this document, it will compile to produce an example of the camera-ready formatting; by leaving it commented out, the document will be anonymized for initial submission. When you first create your submission on softconf, please fill in your submitted paper ID where `***` appears in the `\def\aclpaperid{***}` definition at the top.

The review process is double-blind, so do not include any author information (names, addresses) when submitting a paper for review. However, you should maintain space for names and addresses so that they will fit in the final (accepted) version. The ACL 2016 LaTeX style will create a titlebox space of 2.5in for you when `\aclfinalcopy` is commented out.

2.1 The Ruler

The ACL 2016 style defines a printed ruler which should be presented in the version submitted for review. The ruler is provided in order that reviewers may comment on particular lines in the paper without circumlocution. If you are preparing a document without the provided style files, please arrange for an equivalent ruler to appear on the final output pages. The presence or absence of the ruler should not change the appearance of any other content on the page. The camera ready copy should not contain a ruler. (LaTeX users may uncomment the `\aclfinalcopy` command in the document preamble.)

Reviewers: note that the ruler measurements do not align well with lines in the paper — this turns out to be very difficult to do well when the paper contains many figures and equations, and, when done, looks ugly. Just use fractional references (e.g., the first line on this page is at mark 096.5), although in most cases one would expect that the approximate location will be adequate.

2.2 Electronically-available resources

ACL provides this description in LaTeX2e (`acl2016.tex`) and PDF format (`acl2016.pdf`), along with the LaTeX2e style file used to format it (`acl2016.sty`) and an ACL bibliography style (`acl2016.bst`) and example bibliography (`acl2016.bib`). These files are all available at `acl2016.org/index.php?article_id=9`. We strongly recommend the use of these style files, which have been appropriately tailored for the ACL 2016 proceedings.

2.3 Format of Electronic Manuscript

For the production of the electronic manuscript, you must use Adobe's Portable Document Format (PDF). This format can be generated from postscript files: on Unix systems, you can use `ps2pdf` for this purpose; under Microsoft Windows, you can use Adobe's Distiller, or if you have cygwin

installed, you can use `dvipdf` or `ps2pdf`. Note that some word processing programs generate PDF that may not include all the necessary fonts (esp. tree diagrams, symbols). When you print or create the PDF file, there is usually an option in your printer setup to include none, all, or just non-standard fonts. Please make sure that you select the option of including ALL the fonts. *Before sending it, test your* PDF *by printing it from a computer different from the one where it was created.* Moreover, some word processors may generate very large postscript/PDF files, where each page is rendered as an image. Such images may reproduce poorly. In this case, try alternative ways to obtain the postscript and/or PDF. One way on some systems is to install a driver for a postscript printer, send your document to the printer specifying "Output to a file", then convert the file to PDF.

For reasons of uniformity, Adobe's **Times Roman** font should be used. In LaTeX2e this is accomplished by putting

```
\usepackage{times}
\usepackage{latexsym}
```

in the preamble.

Print-outs of the PDF file on A4 paper should be identical to the hardcopy version. If you cannot meet the above requirements about the production of your electronic submission, please contact the publication chairs above as soon as possible.

2.4 The First Page

Center the title, author name(s) and affiliation(s) across both columns (or, in the case of initial submission, space for the names). Do not use footnotes for affiliations. Use the two-column format only when you begin the abstract.

Title: Place the title centered at the top of the first page, in a 15 point bold font. (For a complete guide to font sizes and styles, see Table 2.) Long titles should be typed on two lines without a blank line intervening. Approximately, put the title at 1in from the top of the page, followed by a blank line, then the author name(s), and the affiliation(s) on the following line. Do not use only initials for given names (middle initials are allowed). Do not format surnames in all capitals (e.g., "Mitchell," not "MITCHELL"). The affiliation should contain the author's complete address, and if possible, an electronic mail address. Leave about 0.75in between the affiliation and the body of the first page.

Abstract: Type the abstract at the beginning of the first column. The width of the abstract text should be smaller than the width of the columns for the text in the body of the paper by about 0.25in on each side. Center the word **Abstract** in a 12 point bold font above the body of the abstract. The abstract should be a concise summary of the general thesis and conclusions of the paper. It should be no longer than 200 words. The abstract text should be in 10 point font.

Text: Begin typing the main body of the text immediately after the abstract, observing the two-column format as shown in the present document. Do not include page numbers in the camera-ready manuscript.

Indent when starting a new paragraph. For reasons of uniformity, use Adobe's **Times Roman** fonts, with 11 points for text and subsection headings, 12 points for section headings and 15 points for the title. If Times Roman is unavailable, use **Computer Modern Roman** (LaTeX2e's default; see section 2.3 above). Note that the latter is about 10% less dense than Adobe's Times Roman font.

Command	Output
{\"a}	ä
{\^e}	ê
{\‘i}	ì
{\.I}	İ
{\o}	ø
{\’u}	ú
{\aa}	å

Command	Output
{\c c}	ç
{\u g}	ğ
{\l}	ł
{\~n}	ñ
{\H o}	ő
{\v r}	ř
{\ss}	ß

Tabel 1: Example commands for accented characters, to be used in, e.g., BibTeX names.

2.5 Sections

Headings: Type and label section and subsection headings in the style shown on the present document. Use numbered sections (Arabic numerals) in order to facilitate cross references. Number subsections with the section number and the subsection number separated by a dot, in Arabic numerals.

Citations: Citations within the text appear in parentheses as [**?**] or, if the author's name appears in the text itself, as Gusfield Gusfield:97. Using the provided LaTeX style, the former is accomplished using `\cite` and the latter with `\shortcite` or `\newcite`. Collapse multiple citations as in [**?**, **?**]; this is accomplished with the provided style using commas within the `\cite` command, e.g., `\cite{Gusfield:97,Aho:72}`. Append lowercase letters to the year in cases of ambiguities. Treat double authors as in [**?**], but write as in [**?**] when more than two authors are involved.

References: We recommend including references in a separate `.bib` file, and include an example file in this release (`naalhlt2016.bib`). Some commands for names with accents are provided for convenience in Table 1. References stored in the separate `.bib` file are inserted into the document using the following commands:

```
\bibliography{acl2016}
\bibliographystyle{acl2016}
```

References should appear under the heading **References** at the end of the document, but before any Appendices, unless the appendices contain references. Arrange the references alphabetically by first author, rather than by order of occurrence in the text. Provide as complete a reference as possible, using a consistent format, such as the one for *Computational Linguistics* or the one in the *Publication Manual of the American Psychological Association* [**?**]. Authors' full names rather than initials are preferred. You may use **standard** abbreviations for conferences[1] and journals[2].

Appendices: Appendices, if any, directly follow the text and the references (but see above). Letter them in sequence and provide an informative title: **Appendix A. Title of Appendix**.

Acknowledgment sections should go as a last (unnumbered) section immediately before the references.

2.6 Footnotes

Footnotes: Put footnotes at the bottom of the page. They may be numbered or referred to by

[1] `https://en.wikipedia.org/wiki/List_of_computer_science_conference_acronyms`

[2] `http://www.abbreviations.com/jas.php`

Type of Text	Font Size	Style
paper title	15 pt	bold
author names	12 pt	bold
author affiliation	12 pt	
the word "Abstract"	12 pt	bold
section titles	12 pt	bold
document text	11 pt	
abstract text	10 pt	
captions	9 pt	
caption label	9 pt	bold
bibliography	10 pt	
footnotes	9 pt	

Tabel 2: Font guide.

asterisks or other symbols.[3] Footnotes should be separated from the text by a line.[4] Footnotes should be in 9 point font.

2.7 Graphics

Illustrations: Place figures, tables, and photographs in the paper near where they are first discussed, rather than at the end, if possible. Wide illustrations may run across both columns and should be placed at the top of a page. Color illustrations are discouraged, unless you have verified that they will be understandable when printed in black ink.

Captions: Provide a caption for every illustration; number each one sequentially in the form: "**Figure 1:** Figure caption.", "**Table 1:** Table caption." Type the captions of the figures and tables below the body, using 9 point text. Table and Figure labels should be bold-faced.

2.8 Accessibility

In an effort to accommodate the color-blind (as well as those printing to paper), grayscale readability for all accepted papers will be encouraged. Color is not forbidden, but authors should ensure that tables and figures do not rely solely on color to convey critical distinctions. Here we give a simple criterion on your colored figures, if your paper has to be printed in black and white, then you must assure that every curves or points in your figures can be still clearly distinguished.

3 Length of Submission

The ACL 2016 main conference accepts submissions of long papers and short papers. Long papers may consist of up to eight (8) pages of content, plus unlimited pages for references. Upon acceptance, final versions of long papers will be given one additional page (up to 9 pages with unlimited pages for references) so that reviewers' comments can be taken into account. Short papers may consist of up to four (4) pages of content, plus unlimited pages for references. Upon acceptance, short papers will be given five (5) pages in the proceedings and unlimited pages for references. For both long

[3]This is how a footnote should appear.
[4]Note the line separating the footnotes from the text.

and short papers, all illustrations and appendices must be accommodated within these page limits, observing the formatting instructions given in the present document. Papers that do not conform to the specified length and formatting requirements are subject to be rejected without review.

4 Double-blind review process

As the reviewing will be blind, the paper must not include the authors' names and affiliations. Furthermore, self-references that reveal the author's identity, e.g., "We previously showed (Smith, 1991) ..." must be avoided. Instead, use citations such as "Smith previously showed (Smith, 1991) ..." Papers that do not conform to these requirements will be rejected without review. In addition, please do not post your submissions on the web until after the review process is complete (in special cases this is permitted: see the multiple submission policy below).

We will reject without review any papers that do not follow the official style guidelines, anonymity conditions and page limits.

5 Multiple Submission Policy

Papers that have been or will be submitted to other meetings or publications must indicate this at submission time. Authors of papers accepted for presentation at ACL 2016 must notify the program chairs by the camera-ready deadline as to whether the paper will be presented. All accepted papers must be presented at the conference to appear in the proceedings. We will not accept for publication or presentation papers that overlap significantly in content or results with papers that will be (or have been) published elsewhere.

Preprint servers such as arXiv.org and ACL-related workshops that do not have published proceedings in the ACL Anthology are not considered archival for purposes of submission. Authors must state in the online submission form the name of the workshop or preprint server and title of the non-archival version. The submitted version should be suitably anonymized and not contain references to the prior non-archival version. Reviewers will be told: "The author(s) have notified us that there exists a non-archival previous version of this paper with significantly overlapping text. We have approved submission under these circumstances, but to preserve the spirit of blind review, the current submission does not reference the non-archival version." Reviewers are free to do what they like with this information.

Authors submitting more than one paper to ACL must ensure that submissions do not overlap significantly ($> 25\%$) with each other in content or results. Authors should not submit short and long versions of papers with substantial overlap in their original contributions.

Acknowledgments

Do not number the acknowledgment section. This section should not be presented for the submission version.

A Supplemental Material

ACL 2016 also encourages the submission of supplementary material to report preprocessing decisions, model parameters, and other details necessary for the replication of the experiments reported in the paper. Seemingly small preprocessing decisions can sometimes make a large difference in performance, so it is crucial to record such decisions to precisely characterize state-of-the-art methods.

Nonetheless, supplementary material should be supplementary (rather than central) to the paper. It may include explanations or details of proofs or deriations that do not fit into the paper, lists of features or feature tempates, sample inputs and outputs for a system, pseudo-code or source code, and data. (Source code and data should be separate uploads, rather than part of the paper).

The paper should not rely on the supplementary material: while the paper may refer to and cite the supplementary material will be available to the reviewers, they will not be asked to review [or even download] the supplementary material.

Appendices (i.e. supplementary material in the form of proofs, tables, or pseudo-code) should come after the references, as shown here. Use `\appendix` before any appendix section to switch the section numbering over to letters.

B Multiple Appendices

. . . can be gotten by using more than one section. We hope you won't need that.